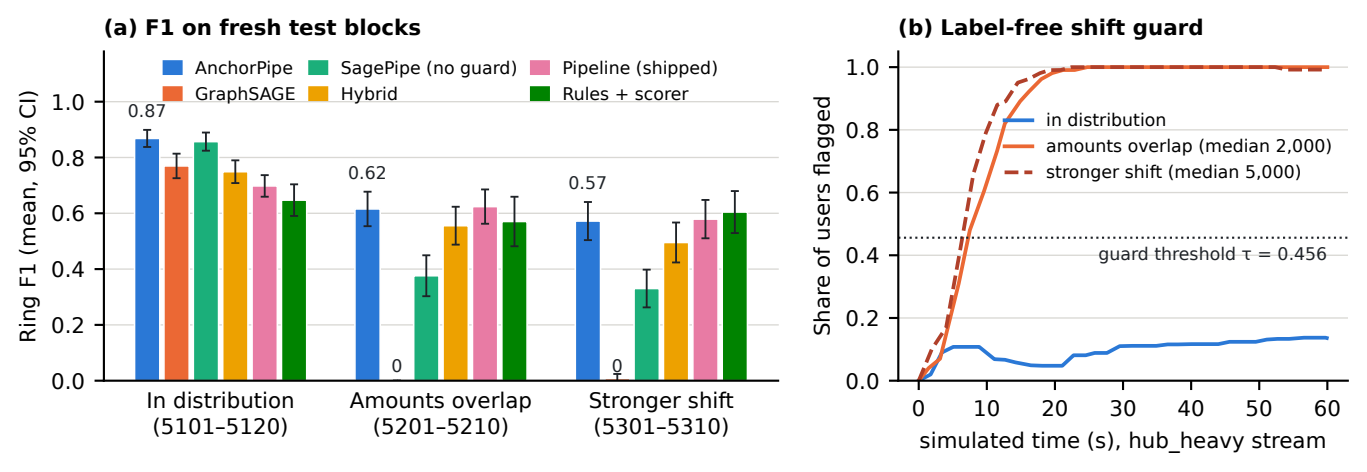
(a) F1 on fresh test blocks
AnchorPipe
GraphSAGE
SagePipe (no guard)
Hybrid
Pipeline (shipped)
Rules + scorer
Ring F1 (mean, 95% CI)
0.87
0.62
0.57
0
0
In distribution
(5101–5120)
Amounts overlap
(5201–5210)
Stronger shift
(5301–5310)
(b) Label-free shift guard
Share of users flagged
in distribution
amounts overlap (median 2,000)
stronger shift (median 5,000)
guard threshold τ = 0.456
simulated time (s), hub_heavy stream